

# Homework 3 — Functions

Session 3 · Domain, codomain, and undoing

**HOW TO ANSWER** Every answer carries one sentence of justification. A value on its own is not a solution — say why it is right. You may discuss problems with anyone and may use AI tools, but you must be able to explain every line you submit.

**BEFORE YOU START** No question here is answered by a verdict alone. "Injective" is not an answer; "injective, because  $f(a_1) = f(a_2)$  forces  $a_1 = a_2$  since..." is. To *disprove* injectivity or surjectivity you need one concrete witness — to *prove* either, you need an argument covering every element.

## Part A · Mechanics

### 1 IMAGES AND PREIMAGES

Let  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  be given by  $f(n) = 3n - 2$ . Compute  $f(\{-1, 0, 2\})$ ,  $f^{-1}(\{7\})$  and  $f^{-1}(\{5\})$ . Explain in one sentence why the third answer is what it is.

**SOLUTION**  $f(\{-1, 0, 2\}) = \{-5, -2, 4\}$ . For  $f^{-1}(\{7\})$ :  $3n - 2 = 7$  gives  $n = 3$ , so the preimage is  $\{3\}$ . For  $f^{-1}(\{5\})$ :  $3n - 2 = 5$  gives  $n = \frac{7}{3}$ , which is not an integer, so no element of the domain maps to 5 and  $f^{-1}(\{5\}) = \emptyset$ . A preimage is allowed to be empty — that is exactly what it means for 5 to lie in the codomain but outside the image.

### 2 IMAGE AGAINST CODOMAIN

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = x^2 - 4x + 7$ . State the image of  $f$  and prove your answer. Is  $f$  surjective? Justify.

**SOLUTION** Complete the square:  $f(x) = (x - 2)^2 + 3$ . Since  $(x - 2)^2 \geq 0$  for every real  $x$ , we get  $f(x) \geq 3$ , so the image is contained in  $[3, \infty)$ . Conversely every  $y \geq 3$  is attained, at  $x = 2 + \sqrt{y - 3}$ . Hence the image is exactly  $[3, \infty)$ .  $f$  is **not** surjective: the codomain is  $\mathbb{R}$ , and 0 (for instance) is never attained because  $f(x) \geq 3 > 0$  for all  $x$ .

Derive each value from the unit circle. For each, state the quadrant and the reference angle before you give the number. A recalled value without that working is not a derivation.

a.  $\cos \frac{2\pi}{3}$

b.  $\sin \frac{3\pi}{4}$

c.  $\cos \frac{7\pi}{6}$

d.  $\sin \frac{5\pi}{3}$

e.  $\tan \frac{3\pi}{4}$

**SOLUTION** (a)  $\frac{2\pi}{3}$  is in quadrant II, reference angle  $\frac{\pi}{3}$ ; cosine is the  $x$ -coordinate, negative there, so  $-\frac{1}{2}$ . (b)  $\frac{3\pi}{4}$ , quadrant II, reference  $\frac{\pi}{4}$ ; sine is the  $y$ -coordinate, positive there, so  $\frac{\sqrt{2}}{2}$ . (c)  $\frac{7\pi}{6}$ , quadrant III, reference  $\frac{\pi}{6}$ ; cosine negative, so  $-\frac{\sqrt{3}}{2}$ . (d)  $\frac{5\pi}{3}$ , quadrant IV, reference  $\frac{\pi}{3}$ ; sine negative, so  $-\frac{\sqrt{3}}{2}$ . (e)  $\tan \frac{3\pi}{4} = \frac{\sin(3\pi/4)}{\cos(3\pi/4)} = \frac{\sqrt{2}/2}{-\sqrt{2}/2} = -1$ .

## 4 CLASSIFICATION

For each function decide whether it is injective, surjective, both or neither. Prove each positive claim; disprove each negative one with an explicit witness.

- a.  $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 5 - 3x$
- b.  $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(n) = n^2 - n$
- c.  $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3 - 3x$
- d.  $f : \mathbb{N} \rightarrow \mathbb{N}, f(n) = n^2$
- e.  $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}, f(x) = \frac{1}{x}$
- f.  $f : \mathbb{R} \rightarrow [-1, 1], f(x) = \sin x$
- g.  $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}, f(a, b) = 2a + 4b$
- h.  $S = \{1, 2, 3\}, f : \mathcal{P}(S) \rightarrow \mathcal{P}(S), f(X) = S \setminus X$

**SOLUTION** (a) **Bijjective.** Injective:  $5 - 3a = 5 - 3b \Rightarrow a = b$ . Surjective: given  $y$ , take  $x = \frac{5-y}{3}$ .

(b) **Neither.** Not injective:  $f(0) = f(1) = 0$ . Not surjective:  $n^2 - n = n(n - 1)$  is a product of consecutive integers, hence even, so the odd value  $1$  is never attained.

(c) **Surjective, not injective.** Not injective:  $f(0) = 0$  and  $f(\sqrt{3}) = 3\sqrt{3} - 3\sqrt{3} = 0$ . Surjective:  $f$  is continuous with  $f(x) \rightarrow \pm\infty$  as  $x \rightarrow \pm\infty$ , so by the intermediate value theorem every real is attained.

(d) **Injective, not surjective.** Injective: for  $a, b \geq 1, a^2 = b^2 \Rightarrow a = b$ . Not surjective:  $2$  is not a perfect square.

(e) **Bijjective.**  $f(f(x)) = x$  on the whole domain, so  $f$  is its own two-sided inverse, and by the theorem a map with a two-sided inverse is a bijection.

(f) **Surjective, not injective.** Not injective:  $\sin 0 = \sin \pi = 0$ . Surjective: the image of  $\sin$  is exactly  $[-1, 1]$ , which here *is* the codomain — note how shrinking the codomain created surjectivity.

(g) **Neither.** Not injective:  $f(2, 0) = 4 = f(0, 1)$ . Not surjective:  $2a + 4b$  is always even, so  $1$  is never attained.

(h) **Bijjective.**  $f(f(X)) = S \setminus (S \setminus X) = X$ , so  $f$  is its own inverse, hence a bijection.

## 5 COMPOSITION

Let  $f : A \rightarrow B$  and  $g : B \rightarrow C$ .

- Prove that if  $g \circ f$  is surjective, then  $g$  is surjective.
- Show by explicit example that  $f$  need *not* be surjective under the same hypothesis.
- In one sentence, say why (a) is about  $g$  and not  $f$ , given that in class we proved  $g \circ f$  injective forces  $f$  injective.

**SOLUTION** (a) Let  $c \in C$ . Since  $g \circ f$  is surjective there is  $a \in A$  with  $g(f(a)) = c$ . Put  $b = f(a) \in B$ ; then  $g(b) = c$ . As  $c$  was arbitrary,  $g$  is surjective.  
 (b)  $A = \{1\}$ ,  $B = \{1, 2\}$ ,  $C = \{1\}$ , with  $f(1) = 1$  and  $g(1) = g(2) = 1$ . Then  $g \circ f$  is surjective onto  $C$ , but  $f$  misses 2.  
 (c) Surjectivity is a statement about what comes *out*, so it is inherited by the last map in the chain; injectivity is a statement about what goes *in*, so it is inherited by the first.

## 6 RESTRICTION

Take the rule  $x \mapsto x^2 - 4x + 7$  from question 2. Choose a domain  $A \subseteq \mathbb{R}$  and a codomain  $B$  making  $f : A \rightarrow B$  a **bijection**, and prove both halves.

**SOLUTION** Take  $A = [2, \infty)$  and  $B = [3, \infty)$ ; write  $f(x) = (x - 2)^2 + 3$ . Injective: for  $x_1, x_2 \geq 2$  the quantities  $x_1 - 2, x_2 - 2$  are  $\geq 0$ , and  $(x_1 - 2)^2 = (x_2 - 2)^2$  with both bases non-negative gives  $x_1 - 2 = x_2 - 2$ . Surjective: given  $y \geq 3$ , put  $x = 2 + \sqrt{y - 3}$ , which lies in  $A$  and satisfies  $f(x) = y$ . ( $A = (-\infty, 2]$  with the same  $B$  works equally well.)

The following was produced by a chatbot. It contains **three** distinct errors. Find all three; for each, say what is wrong and give a witness where one is needed.

**Claim.**  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = \frac{x}{x^2 + 1}$ , is a bijection, and its inverse is  $f^{-1}(x) = \frac{x^2 + 1}{x}$ .

**"Proof."**  $f$  is injective because it is a ratio of polynomials with no repeated factors. It is surjective because its domain and codomain are both  $\mathbb{R}$ . Inverting a fraction gives the inverse function.

**SOLUTION** **Error 1 – not injective.** "Ratio of polynomials with no repeated factors" is not a criterion for anything. Witness:  $f(2) = \frac{2}{5}$  and  $f(\frac{1}{2}) = \frac{1/2}{5/4} = \frac{2}{5}$ , so  $f(2) = f(\frac{1}{2})$  with  $2 \neq \frac{1}{2}$ .

**Error 2 – not surjective.** Domain and codomain being equal says nothing about surjectivity. In fact  $(x \mp 1)^2 \geq 0$  gives  $x^2 + 1 \geq \pm 2x$ , so  $|f(x)| \leq \frac{1}{2}$  for every  $x$ ; the image is  $[-\frac{1}{2}, \frac{1}{2}]$  and 1 is never attained.

**Error 3 – inverse against reciprocal.**  $\frac{x^2+1}{x}$  is  $\frac{1}{f(x)}$ , not  $f^{-1}$ . Undoing a function is unrelated to flipping a fraction:  $f^{-1}$  satisfies  $f^{-1}(f(x)) = x$ , and here  $\frac{1}{f(f(x))} \neq x$  in general. Since  $f$  is not a bijection, no inverse function exists at all.

Each sentence is sloppy or false as written. Rewrite each so that it is unambiguous and true.

- " $x^2$  is not injective."
- "Every function has an inverse — you just swap  $x$  and  $y$ ."
- " $\sin^{-1}(x)$  means  $\frac{1}{\sin x}$ ."

**WHAT A GOOD ANSWER DOES**

(a) A rule is not a function: e.g. "the function  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = x^2$ , is not injective, since  $f(-1) = f(1)$  but  $-1 \neq 1$ ." Any answer that supplies a domain and codomain and a witness is acceptable.

(b) "A function has an inverse if and only if it is a bijection. Swapping  $x$  and  $y$  in the equation of the graph yields a function precisely in that case."

(c) " $\sin^{-1}$  denotes the inverse of  $\sin$  restricted to  $[-\frac{\pi}{2}, \frac{\pi}{2}]$ , where it is a bijection onto  $[-1, 1]$ . The reciprocal  $\frac{1}{\sin x}$  is a different function, written  $\csc x$ ."